

Пункт 1

```
eltex-v25@eltex-2026-summer:~$ ssh-keygen -t rsa -b 4096 -f ~/.ssh/id_rsaaa -N ""
Generating public/private rsa key pair.
Created directory '/home/eltex-v25/.ssh'.
Your identification has been saved in /home/eltex-v25/.ssh/id_rsaaa
Your public key has been saved in /home/eltex-v25/.ssh/id_rsaaa.pub
The key fingerprint is:
SHA256:kS0Ip3lykdjKD2oQFjf96SVuhnQ07JEdykOkCk5C9bY eltex-v25@eltex-2026-summer
The key's randomart image is:
+---[RSA 4096]-----+
| oo+oo=+o.. |
| o.. +**O.+ |
| o+ .+=B+B . |
| = . *=.*.+ |
| o o.E= S |
| o ..= |
| . o |
| |
| |
+----[SHA256]-----+
```

Пункт 2

```
eltex-v25@eltex-2026-summer:~$ ssh-copy-id root@172.16.9.199
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/eltex-v25/.ssh/id_rsaaa.pub"
The authenticity of host '172.16.9.199 (172.16.9.199)' can't be established.
ED25519 key fingerprint is SHA256:YNVzj+RtetZfXv5XUn/CUjeEcaMD4eg3jy8EuGGIS+o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@172.16.9.199's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'root@172.16.9.199'"
and check to make sure that only the key(s) you wanted were added.
```

Пункт 3

```
eltex-v25@eltex-2026-summer:~$ ssh root@172.16.9.199
root@172.16.9.199's password:
root@eltex-practice2-v25:~# cat ~/.ssh/authorized_keys
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQCAQC3q9x6WFDBGfLfYwFX4GBwtOP8OryKLfXGOt07shO9cmkAD
XxUu0ON3VIPdyL4FVBQBbHy2vP9vVaS/Ho1uDI1YdKjxf2/nGqFiHmtwG+/4qg0zXBp3oN5IpNyF6oogFLFXWb
L3f5hnwCtrlAPxL8bfIPzuj/ekplwB6OfCQRrmcUUnjsDdQNKnhTtPIKXu91FujMc2EyrDedqzCo6PgbPkySU/4TJb
W0gSL094RC1vr6jG+r/F7pRHObFTkXXxTJQz+lnx3CfycU2hwmToCk7rB6tlk3l68gY1JWNPPyJvTkGXy5H13
vIMJtFuF4jjjQU7/xqCzFdO2B63EAHV4IfPTw8do6MGZdtVYAhTWOZm1nRP1c/KfLaVNiHKVzwgM054ZQ/SZ
CZyDXdA0XA/B0RkdY5bjswBz3DDYyQzdklsNzFYENGKsrOYL6kqtuH+WLVGx+1fo2MUwVGirw6wyGaisK
MXHtPg+3AKw+CZ4F/DL2fWNA+I0q2SQdA8h59auRYEGnX39+t6zSBpnP9J9HkFoHCeAKv+Uxaa6Kpvmmbt
```

```
MZA93TqgdICeHDBO1v1J/rI28Ljr/DreGTuDL+Ti7rqc8LXrtliHGuHq9cvnKI7Hqp1mYmjNKqfN6WeWpXV5u/Wfmc5UMDlxRF4h/VpMOxqXh9UcT0NUlgwAM4eQ== eltex-v25@eltex-2026-summer
```

Пункт 4

```
root@eltex-practice2-v25:~# useradd -m -s /bin/bash user1
root@eltex-practice2-v25:~# passwd user1
New password:
Retype new password:
passwd: password updated successfully
```

Пункт 5

```
root@eltex-practice2-v25:~# adduser user2
info: Adding user `user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user2' (1002) ...
info: Adding new user `user2' (1002) with group `user2 (1002)' ...
info: Creating home directory `/home/user2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
No password has been supplied.
New password:
Retype new password:
No password has been supplied.
New password:
Retype new password:
No password has been supplied.
passwd: Authentication token manipulation error
passwd: password unchanged
Try again? [y/N] y
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n]
info: Adding new user `user2' to supplemental / extra groups `users' ...
info: Adding user `user2' to group `users' ...
root@eltex-practice2-v25:~# adduser user3
info: Adding user `user3' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user3' (1003) ...
info: Adding new user `user3' (1003) with group `user3 (1003)' ...
elinfo: Creating home directory `/home/user3' ...
info: Copying files from `/etc/skel' ...
```

```
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
Try again? [y/N]
Changing the user information for user3
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n]
info: Adding new user `user3' to supplemental / extra groups `users' ...
info: Adding user `user3' to group `users' ...
```

Пункт 6

```
root@eltex-practice2-v25:~# usermod -s /usr/sbin/nologin user3
root@eltex-practice2-v25:~# su user3
This account is currently not available.
root@eltex-practice2-v25:~# su -s /bin/bash user3
user3@eltex-practice2-v25:/root$ exit
exit
```

Пункт 7

```
root@eltex-practice2-v25:~# groupadd sharegroup
root@eltex-practice2-v25:~# usermod -aG sharegroup user1
root@eltex-practice2-v25:~# usermod -aG sharegroup user2
root@eltex-practice2-v25:~# usermod -aG sharegroup user3
root@eltex-practice2-v25:~# id user1
uid=1001(user1) gid=1001(user1) groups=1001(user1),1004(sharegroup)
root@eltex-practice2-v25:~# id user2
uid=1002(user2) gid=1002(user2) groups=1002(user2),100(users),1004(sharegroup)
root@eltex-practice2-v25:~# id user3
uid=1003(user3) gid=1003(user3) groups=1003(user3),100(users),1004(sharegroup)
```

Пункт 9

```
root@eltex-practice2-v25:~# mkdir -p /opt/share
root@eltex-practice2-v25:~# chown :sharegroup /opt/share
root@eltex-practice2-v25:~# chmod 2770 /opt/share
```

Пункт 10

```
root@eltex-practice2-v25:~# echo "umask 027" >> /home/user1/.bashrc
root@eltex-practice2-v25:~# chown user1:user1 /home/user1/.bashrc
```

Пункт 11

```
root@eltex-practice2-v25:~# su - user1
user1@eltex-practice2-v25:~$ touch /opt/share/file_user1.txt
user1@eltex-practice2-v25:~$ exit
logout
root@eltex-practice2-v25:~# su - user2
user2@eltex-practice2-v25:~$ touch /opt/share/file_user2.txt
user2@eltex-practice2-v25:~$ rm /opt/share/file_user1.txt
rm: remove write-protected regular empty file '/opt/share/file_user1.txt'? yes
user2@eltex-practice2-v25:~$ exit
logout
```

В Linux права на удаление файла определяются правами записи на родительский каталог, а не на сам файл. Так как у группы есть права `w` на папку `/opt/share`, любой член группы может удалять файлы других участников

Пункт 12

```
root@eltex-practice2-v25:~# chmod +t /opt/share
root@eltex-practice2-v25:~# su - user1
user1@eltex-practice2-v25:~$ touch /opt/share/file_user1.txt
user1@eltex-practice2-v25:~$ exit
logout
root@eltex-practice2-v25:~# su - user2
user2@eltex-practice2-v25:~$ rm /opt/share/file_user1.txt
rm: remove write-protected regular empty file '/opt/share/file_user1.txt'? yes
rm: cannot remove '/opt/share/file_user1.txt': Operation not permitted
```

Бит Sticky Bit (+t) защищает файлы внутри каталога от удаления сторонними пользователями. Удалить или переименовать файл теперь может только владелец этого файла, владелец каталога или `root`.

Пункт 13

```
root@eltex-practice2-v25:~# which dmesg
/usr/bin/dmesg
root@eltex-practice2-v25:~# echo "user1 ALL=(ALL) NOPASSWD: /usr/bin/dmesg" > /etc/sudoers.d/user1
root@eltex-practice2-v25:~# cat << 'EOF' > /usr/local/bin/run_dmesg
#!/bin/bash
/usr/bin/dmesg
EOF
root@eltex-practice2-v25:~# cat /usr/local/bin/run_dmesg
#!/bin/bash
/usr/bin/dmesg
```

```
root@eltex-practice2-v25:~# chmod 4755 /usr/local/bin/run_dmesg
```

Пункт 14

```
root@eltex-practice2-v25:~# chage -M 10 user1
root@eltex-practice2-v25:~# chage -M 10 user2
root@eltex-practice2-v25:~# chage -M 10 user3
```

Пункт 15

```
root@eltex-practice2-v25:~# echo "Михайлов Андрей" > /etc/motd
```

Пункт 16

```
root@eltex-practice2-v25:~# mkdir -p /root/etc_backup
root@eltex-practice2-v25:~# rsync -av /etc/ /root/etc_backup/
...
xml/xml-core.xml.old

sent 2,415,269 bytes received 19,236 bytes 1,623,003.33 bytes/sec
total size is 2,332,068 speedup is 0.96
```

Пункт 17

```
root@eltex-practice2-v25:~# tar -czf /root/etc_backup.tar.gz -C /root etc_backup
root@eltex-practice2-v25:~# tar -cjf /root/etc_backup.tar.bz2 -C /root etc_backup
root@eltex-practice2-v25:~# tar -cf /root/etc_backup.tar -C /root etc_backup
root@eltex-practice2-v25:~# 7z a /root/etc_backup.tar.7z /root/etc_backup.tar
```

```
7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
64-bit locale=en_US.UTF-8 Threads:2 OPEN_MAX:1024
```

```
Scanning the drive:
1 file, 3420160 bytes (3340 KiB)
```

```
Creating archive: /root/etc_backup.tar.7z
```

```
Add new data to archive: 1 file, 3420160 bytes (3340 KiB)
```

```
Files read from disk: 1
Archive size: 466014 bytes (456 KiB)
```

```
Everything is Ok
root@eltex-practice2-v25:~# rm /root/etc_backup.tar
root@eltex-practice2-v25:~# ls -lh /root/etc_backup.tar.gz /root/etc_backup.tar.bz2 /root/etc_backup.tar.7z
-rw-r--r-- 1 root root 456K Aug 31 03:30 /root/etc_backup.tar.7z
-rw-r--r-- 1 root root 527K Aug 31 03:28 /root/etc_backup.tar.bz2
-rw-r--r-- 1 root root 564K Aug 31 03:26 /root/etc_backup.tar.gz
```

Пункт 18

```
root@eltex-practice2-v25:~# gunzip /root/etc_backup.tar.gz
root@eltex-practice2-v25:~# tar -rf /root/etc_backup.tar -C /root etc_backup/motd
root@eltex-practice2-v25:~# gzip /root/etc_backup.tar
```

Пункт 19

```
root@eltex-practice2-v25:~# tar -djf /root/etc_backup.tar.bz2 -C /root
etc_backup/motd: Mod time differs
etc_backup/motd: Size differs
```

Пункт 20

```
root@eltex-practice2-v25:~# mkdir -p /root/etc_backup_gzip /root/etc_backup_7zip
root@eltex-practice2-v25:~# tar -xzf /root/etc_backup.tar.gz -C /root/etc_backup_gzip
root@eltex-practice2-v25:~# 7z x -o/root/etc_backup_7zip /root/etc_backup.tar.7z
tar -xf /root/etc_backup_7zip/etc_backup.tar -C /root/etc_backup_7zip
```

7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
64-bit locale=en_US.UTF-8 Threads:2 OPEN_MAX:1024

Scanning the drive for archives:
1 file, 466014 bytes (456 KiB)

Extracting archive: /root/etc_backup.tar.7z

--

Path = /root/etc_backup.tar.7z

Type = 7z

Physical Size = 466014

Headers Size = 138

Method = LZMA2:22

Solid = -

Blocks = 1

Everything is Ok

Size: 3420160

Compressed: 466014

```
root@eltex-practice2-v25:~# diff /root/etc_backup_gzip/etc_backup/etc/motd
```

```
/root/etc_backup_7zip/etc_backup/etc/motd
```

```
diff: /root/etc_backup_gzip/etc_backup/etc/motd: No such file or directory
```

```
diff: /root/etc_backup_7zip/etc_backup/etc/motd: No such file or directory
root@eltex-practice2-v25:~# diff /root/etc_backup_gzip/etc_backup/etc/motd
/root/etc_backup_7zip/etc_backup/motd
diff: /root/etc_backup_gzip/etc_backup/etc/motd: No such file or directory
root@eltex-practice2-v25:~# diff /root/etc_backup_gzip/etc_backup/motd /root/etc_backup_7zip/etc_backup/motd
1c1
< 2026-08-31 03:31:06
---
> Михайлов Андрей
```